

Laplace Transformation

4. Laplace transform of integral:

$$L\left[\int_0^t f(\tau) d\tau\right] = \frac{1}{s} F(s), \text{ if } L\{f(t)\} = F(s)$$

Laplace transformation of unit step function:

$$u(t-a) = \begin{cases} 1 & t \geq a \\ 0 & t < a \end{cases}$$

$$L\{u(t-a)\} = \int_0^{\infty} e^{-st} u(t-a) dt = \int_0^a e^{-st} \cdot 0 dt + \int_a^{\infty} e^{-st} \cdot 1 dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_a^{\infty} = \frac{-1}{s} (e^{-\infty} - e^{-sa})$$

$$L\{u(t-a)\} = \frac{e^{-sa}}{s}$$

Question: $u(t-t_0) = \begin{cases} 1 & t \geq t_0 \\ 0 & t < t_0 \end{cases}$

Solution: $L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt = \int_0^{t_0} e^{-st} \cdot 0 dt + \int_{t_0}^{\infty} e^{-st} \cdot 1 dt$

$$= \left[\frac{e^{-st}}{-s} \right]_{t_0}^{\infty} = \frac{-1}{s} [0 - e^{-st_0}] = \frac{e^{-st_0}}{s}$$

Question: $L\{f(t)\} = L\{u(t-a)\} - L\{u(t-b)\}$

Solution: $L\{f(t)\} = \frac{e^{-as}}{s} - \frac{e^{-bs}}{s}$

